

WAFT Architecture

Project: System Design

Version: 1.0

System Architecture

WAFT is built on three core systems working in harmony.

Core Components

1. Template System

12 diverse document generators covering academic, business, technical, and creative formats.

2. Reflection System

Uses AST analysis to observe the codebase and identify documentation needs.

3. Binder System

Assembles multiple documents into cohesive collections with covers, TOCs, and dividers.

Data Flow

```
User Request
  ↓
Template Selection
  ↓
Content Generation
  ↓
PDF Rendering (WeasyPrint)
  ↓
Document Assembly (Binder)
  ↓
Final Output
```

Technology Stack

- **WeasyPrint** - HTML/CSS to PDF conversion
- **Jinja2** - Template engine
- **pypdf** - PDF manipulation
- **AST** - Python code analysis